

NAVINESH P K

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OBJECTIVE

I am an enthusiastic Electronics and Communication Engineering student with a strong passion for embedded systems and PCB design. I enjoy creating innovative and reliable hardware-software solutions that address real-world challenges. Skilled in microcontroller programming, circuit design, and PCB layout using tools like KiCad and Altium, I strive for precision and efficiency in every project. I am dedicated to continuous learning and staying updated with emerging technologies. Eager to contribute to a forward-thinking team, I bring a collaborative spirit and a problem-solving mindset to every opportunity.

EDUCATION

B.E. Electronics and Communication Engineering

2023-2027

- K.S. Rangasamy College of Technology, Tiruchengode
- CGPA: 7.9/10 (up to 5th Semester)

Higher Secondary Certificate (HSC)

2021-2023

- Sri S.R.V Matric.Hr.sec.School, Narasingapuram, Attur.
- Percentage: 71.5%

Secondary School Leaving Certificate (SSLC)

2021

- Sri S.R.V Matric.Hr.sec.School, Narasingapuram, Attur.
- PASS

SKILLS AND PROFICIENCIES

- | | | |
|-------------------|-----------|---------------------|
| • Cadence Allegro | • Ki-Cad | • Bare metal coding |
| • Vivado | • AutoCAD | • Embedded C |

ONLINE CERTIFICATE COMPLETED

- English for Competitive Exams by NPTEL (Jan - Apr 2024)
- Introduction to Industrial IoT by NPTEL (Jan - Apr 2025)
- Circuit Simulation Onramp in MathWorks
- Infosys springboard (Data Structures and Algorithms, Building an Agile Mindset . Site Reliability: Engineering)

MEMBERSHIP

- The Institution of Electronics and Telecommunication Engineers (IETE).

INTERNSHIP

- Undergone two month (01.04.2025 - 01.06.2025) Embedded System Developer Virtual Internship at Microchip Technology, USA.
- Undergone 15 days Offline Internship (31.12.2025 – 14.01.2026) at SSD Trace Tech PCB Private Limited at Coiminator.

PROJECT

- **Currently Working:** IoT Backend-as-a-Service (BaaS) Platform: Designed and developed a fully self-hosted, multi-tenant IoT Backend-as-a-Service platform supporting MQTT, HTTP, BLE, and custom LoRa communication protocols. Implemented real-time telemetry processing, WebSocket monitoring, event-driven microservices architecture, device authentication, and PostgreSQL-based scalable data management using Dockerized deployment.
- Vision-Enhanced Driver Safety System: Designed an AI-IoT solution using Raspberry Pi, OpenCV, and Firebase to detect driver drowsiness, alcohol, and accidents with SMS, voice alerts, GPS tracking, and engine cut-off.
- Automatic Fire Extinguisher: Created an automated fire detection and extinguishing system that activates upon sensing flames, enhancing safety in residential or commercial spaces.
- Embedded Motor Speed Control System: Developed a DC motor speed control setup using electronic control methods and voltage regulation techniques. The project explored different approaches for controlling motor speed without using conventional potentiometer-based methods.
- Wireless RF Transmitter and Receiver Circuit: Implemented a wireless communication system using RF technology to transmit and receive data over a short range, useful for remote control applications.
- Shot Clock Using 2 Digit Seven Segment Display: Developed a countdown timer for basketball games, displayed on a two-digit seven-segment display, to ensure fair play and adherence to time regulations.
- Temperature Controlled DC Fan Circuit: Designed a circuit that automatically adjusts the speed of a DC fan based on the ambient temperature, providing energy-efficient cooling.

SOFT SKILLS

- Coordinated and managed 20+ technical and non-technical college events with strong leadership and stage management skills.
- Strong organizational, teamwork, and problem-solving abilities demonstrated through hackathons, workshops, and student activities
- Effective communication skills in both Tamil and English with confidence in public speaking and teamwork.
- Skilled in Canva design and creative content preparation for presentations, certificates, posters, and event materials.

EXTRA CURRICULAR ACTIVITIES

- Vice President of SMV (service moto volunteer).
- Student Coordinator of NSS.

ACHIEVEMENTS

- Participated in a National-Level Technical Symposium organized by [Coimbatore Institute of Technology].
- Presented a paper at One Day National-Level Technical Symposium organized by [Sree Sakthi Engineering College, Coimbatore].
- SRM Smart Park Hackathon – Exhibition of Working Models Participated in the SRM Smart Park Hackathon 2025 at SRM Institute of Science and Technology, Kattankulathur. Worked with the “Techtitans” team to develop and present an innovative working model, showcasing teamwork, problem-solving, and technical presentation skills.

AREA OF INTREST

- Embedded systems
- IOT & Hardware
- Bare metal coding
- C Programming

PERSONAL DETAILS

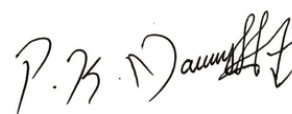
Date of Birth & Age - 06.09.2005 & 20
Parents' Name - Mr. P. Pachamuthu & Mrs. K. Kayalvizhi
Personal Traits - Leadership, Conscientiousness and Agreeableness
Languages Known - Tamil & English (Read, Write and Speak)
Hobby - Movies, Reading, Poetry and Listening to Music

Declaration

I hereby declare that all the information provided above is true to the best of my knowledge and responsibility for the correctness of the above-mentioned particulars.

Date : 11/05/2026

Place : Tiruchengode


(NAVINESH P K)